

Remaining qualitative boundaries

Matched test images and generated answer excerpts; blue text marks the guided student.



Fine-grained semantic failure

Local visual evidence is not enough for full scene semantics.

Observed preference: The teacher remains preferred even though the guided student identifies the skull.

Teacher VLM

"The image shows a person standing next to a large, detailed dinosaur skull mounted on a stand. The skull appears to be that of a Triceratops, characterized by its three prominent horns and frill. The person is ..."

Student, explanation-guided

"The image shows a close-up of a dinosaur skull, specifically that of a Triceratops. The skull is displayed in a museum setting, with a window in the background allowing natural light to illuminate the ..."

Reading The student captures the Triceratops skull but misses broader context, including the nearby person and richer scene description.



Generation robustness failure

The plain MSE visual swap can break generation outright.

Observed preference: The baseline produces a no-image response; guided variants mostly avoid this failure mode.

Student, global MSE

"I'm sorry, but I cannot see any image attached to your message. Could you please provide more information or rephrase your question?"

Student, explanation-guided

"The image depicts a dinosaur skeleton on display in what appears to be a museum or exhibition hall. The skeleton is mounted on a stand and is positioned in a standing posture, with its legs spread apart and arms ..."

Reading Explanation guidance improves visual-path usability: the guided answer describes the skeleton, while the MSE student says it cannot see the image.